

DCs API

Summary

Resource	Operation	Description
DC	GET /dcs	Get all DC

Details

GET /dcs

- Status Codes:
- [200 OK](#) – The [DC](#) objects are returned.
 - [401 Unauthorized](#) – API key is invalid.

Example response

```
{
  "status": "success",
  "data": {
    "dcs": [{
      "id": "the dc's id",
      "name": "the dc's name",
      "time_zone": "America/Los_Angeles"
    }]
  }
}
```

Drivers API

Summary

Resource	Operation	Description
Drivers	GET /dcs/(dc_id)/drivers/(driver_id)/driver_bulk_info	Get driver metadata and location.
	PATCH /dcs/(dc_id)/drivers/(driver_id)/finish-routes	Finish all active routes assigned to this
	PUT /dcs/(dc_id)/drivers/(driver_id)/finish-routes	
	GET /dcs/(dc_id)/drivers/(driver_id)/active-routes	Get all active routes assigned to this dri
	GET /dcs/(dc_id)/drivers/(driver_id)/location	Get the latest known location of this di
	GET /dcs/(dc_id)/drivers/(driver_id)	Get a driver
	PATCH /dcs/(dc_id)/drivers/(driver_id)	Update a driver
	DELETE /dcs/(dc_id)/drivers/(driver_id)	Delete a driver
	POST /dcs/(dc_id)/drivers	Create a driver
	GET /dcs/(dc_id)/drivers	Get all drivers for the dc

 

Details

GET [/dcs/\(dc_id\)/drivers/\(driver_id\)/driver_bulk_info](#)

- Parameters:
- `dc_id` (*string*) – The dc id
 - `driver_id` (*string*) – The driver's id

- Status Codes:
- [200 OK](#) –

`ref`: The latest known location of this driver is returned.

- [401 Unauthorized](#) – API key is invalid.
- [403 Forbidden](#) – [DC](#) does not exist or user does not have permission to access it.
- [404 Not Found](#) – [Driver](#) does not exist.

Example response

```
{
  "status": "success",
  "data": {
    "driver": {
      "id": "the driver's id",
      "login_token": "secret-login-token",
      "name": "the driver's name"
    }
    "driver_location": {
      "timestamp": 1567702800000,
      "location": {
        "latitude": 37.7749,
        "longitude": -122.4194
      }
    }
  }
}
```

Please note that if the driver exists but has not reported any locations to Foxtrot, the *driver_location* object may be null.

PATCH `/dcs/(<dc_id>)/drivers/(<driver_id>)/finish-routes`

- Parameters:
- **dc_id** (*string*) – The dc id
 - **driver_id** (*string*) – The driver's id

- Status Codes:
- **200 OK** – [Routes](#) were finalized.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – [DC](#) does not exist or user does not have permission to access it.
 - **404 Not Found** – [Driver](#) does not exist.

PUT /dcs/([dc_id](#))/drivers/([driver_id](#))/finish-routes

- Parameters:
- **dc_id** (*string*) – The dc id
 - **driver_id** (*string*) – The driver's id

- Status Codes:
- **200 OK** – [Routes](#) were finalized.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – [DC](#) does not exist or user does not have permission to access it.
 - **404 Not Found** – [Driver](#) does not exist.

GET /dcs/([dc_id](#))/drivers/([driver_id](#))/active-routes

- Parameters:
- **dc_id** (*string*) – The dc id
 - **driver_id** (*string*) – The driver's id

Status Codes: • 200 OK –

ref: List of active routes for driver is returned.

- 401 Unauthorized – API key is invalid.
- 403 Forbidden – DC does not exist or user does not have permission to access it.
- 404 Not Found – Driver does not exist.

Example response

```
{
  "status": "success",
  "data": {
    "active_routes": [{
      "planned_start_time": 1567702800000,
      "route_id": "route id",
    }, {
      "planned_start_time": 1567731600000,
      "route_id": "route id 2",
    }
  ]
}
```

GET /dcs/(dc_id)/drivers/(driver_id)/location

Parameters: • dc_id (*string*) – The dc id

• driver_id (*string*) – The driver's id

Status Codes: • 200 OK –

ref: The latest known location of this driver is returned.

- 401 Unauthorized – API key is invalid.
- 403 Forbidden – DC does not exist or user does not have permission to access it.
- 404 Not Found – Driver does not exist.

Example response

```
{
  "status": "success",
  "data": {
    "driver_location": {
      "timestamp": 1567702800000,
      "location": {
        "latitude": 37.7749,
        "longitude": -122.4194
      }
    }
  }
}
```

Please note that if the driver exists but has not reported any locations to Foxtrot, the *driver_location* object may be null.

GET /dcs/(*dc_id*)/drivers/(*driver_id*)

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **driver_id** (*string*) – The driver's id

- Status Codes:**
- **200 OK** – **Driver** object is returned.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.
 - **404 Not Found** – **Driver** does not exist.

Example response

```
{
  "status": "success",
  "data": {
    "driver": {
      "id": "the driver's id",
      "login_token": "secret-login-token",
      "name": "the driver's name"
    }
  }
}
```

PATCH /dcs/*(dc_id)*/drivers/*(driver_id)*

- Parameters:
- **dc_id** (*string*) – The dc id
 - **driver_id** (*string*) – The driver's id

Request JSON Object:

- **driver_name** (*string*) – The driver's new name

- Status Codes:
- **200 OK** – **Driver** was updated.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.
 - **404 Not Found** – **Driver** does not exist.

Example request

```
{
  "driver_name": "John's new name"
}
```

DELETE /dcs/*(dc_id)*/drivers/*(driver_id)*

- Parameters:
- **dc_id** (*string*) – The dc id
 - **driver_id** (*string*) – The driver's id

- Status Codes:
- **200 OK** – **Driver** was deleted.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.
 - **404 Not Found** – **Driver** does not exist.

POST /dcs/(**dc_id**)/drivers

- Parameters:
- **dc_id** (*string*) – The dc id

Request JSON Object:

- **driver_name** (*string*) – The driver's name
- **driver_id** (*string*) – The driver's id

- Status Codes:
- **200 OK** – **Driver** was created.
 - **400 Bad Request** – Check error code within the response.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.

Error Code	Description
2000	Driver with that id already exists.

Example request


```
{
  "driver_name": "John",
  "login_token": "secret-login-token",
  "driver_id": "John's id"
}
```

GET /dcs/*(dc_id)*/drivers

Parameters: • **dc_id** (*string*) – The dc id

Status Codes: • **200 OK** – **Driver** objects are returned.

- **401 Unauthorized** – API key is invalid.
- **403 Forbidden** – **DC** does not exist or user does not have permission to access it.
- **404 Not Found** – **Driver** does not exist.

Example response

```
{
  "status": "success",
  "data": {
    "drivers": [{
      "id": "the driver's id",
      "login_token": "secret-login-token",
      "name": "the driver's name"
    }, {
      "id": "the driver's id 2",
      "login_token": "secret-login-token",
      "name": "the driver's name 2"
    }
  ]
}
```

Routes API

Summary

Resource	Operation
Routes	POST /dcs/{dc_id}/routes/{route_id}/waypoints/{waypoint_id}/deliveries/authorize-reattempts
	POST /dcs/{dc_id}/routes/{route_id}/waypoints/{waypoint_id}/deliveries
	GET /dcs/{dc_id}/routes/{route_id}/waypoints/{waypoint_id}/deliveries
	GET /dcs/{dc_id}/routes/import-status/{txid}
	GET /dcs/{dc_id}/routes/find_by_date/{date}
	PATCH /dcs/{dc_id}/routes/{route_id}/vehicle-assignment
	PUT /dcs/{dc_id}/routes/{route_id}/vehicle-assignment
	PATCH /dcs/{dc_id}/routes/{route_id}/driver-assignment
	PUT /dcs/{dc_id}/routes/{route_id}/driver-assignment
	DELETE /dcs/{dc_id}/routes/{route_id}/driver-assignment
	GET /dcs/{dc_id}/routes/{route_id}/completion-time
	PATCH /dcs/{dc_id}/routes/{route_id}/deactivate
	PUT /dcs/{dc_id}/routes/{route_id}/deactivate
	GET /dcs/{dc_id}/routes/{route_id}/waypoints/{waypoint_id}

Resource	Operation
	POST /dcs/(dc_id)/routes/(route_id)/waypoints
	GET /dcs/(dc_id)/routes/(route_id)/waypoints
	PATCH /dcs/(dc_id)/routes/(route_id)/activate
	PUT /dcs/(dc_id)/routes/(route_id)/activate
	PATCH /dcs/(dc_id)/routes/(route_id)/finish
	PUT /dcs/(dc_id)/routes/(route_id)/finish
	PATCH /dcs/(dc_id)/routes/(route_id)/start
	PUT /dcs/(dc_id)/routes/(route_id)/start
	DELETE /dcs/(dc_id)/routes/(route_id)
	GET /dcs/(dc_id)/routes/(route_id)
	POST /dcs/(dc_id)/routes

Details

POST /dcs/([dc_id](#))/routes/([route_id](#))/waypoints/([waypoint_id](#))/deliveries/authorize-reattempts

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id
 - **waypoint_id** (*string*) – The waypoint id

- Status Codes:**
- **200 OK** – :ref:` The deliveries are marked for reattempt
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – [DC](#) does not exist or user does not have permission to access it.
 - **404 Not Found** – [Route](#), [Waypoint](#), or one of the provided [Deliveries](#) does not exist.

Status 400 error codes

Error Code	Description
3012	The route has already been finished (so more deliveries cannot be added).

Example request

```
{
  "deliveries": [
    {
      "id": "an optional unique identifier for the reattempt",
      "delivery_id": "delivery_id_1234",
      "quantity": 33.0
    },
    {
      "delivery_id": "delivery_id_5678"
    }
  ],
  "time_window": {
    "start": 1000,
    "end": 5000
  },
  "notes": "An optional note to include"
}
```

POST /dcs/([dc_id](#))/routes/([route_id](#))/waypoints/([waypoint_id](#))/deliveries

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id
 - **waypoint_id** (*string*) – The waypoint id

- Status Codes:**
- [200 OK](#) – Delivery was added.
 - [401 Unauthorized](#) – API key is invalid.
 - [403 Forbidden](#) – [DC](#) does not exist or user does not have permission to access it.
 - [404 Not Found](#) – [Route](#) or [Waypoint](#) does not exist.

Status 400 error codes

Error Code	Description
3007	Cannot import another delivery with the same id.
3012	The route has already been finished (so more deliveries cannot be added).

Example request

```
{
  "deliveries": [
    {
      "product": "product",
      "quantity": 33.0,
      "id": "globally unique delivery id"
    },
    {
      "product": "another product",
      "quantity": 53.0,
      "id": "another globally unique delivery id"
    }
  ]
}
```

GET /dcs/(*dc_id*)/routes/(*route_id*)/waypoints/(*waypoint_id*)/deliveries

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id
 - **waypoint_id** (*string*) – The waypoint id

- Status Codes:**
- **200 OK** – Delivery objects are returned.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.
 - **404 Not Found** – **Route** or **Waypoint** does not exist.

Example Responses

```
{
  "deliveries": [{
    "id": "delivery-id-1",
    "name": "delivery-name-1",
    "quantity": 7,
    "attempts": [{
      "attempt_status": "SUCCESSFUL",
      "timestamp": 1570561955123,
      "driver_notes": "Product Delivered"
    }, ...],
    "delivery_code": "AF207",
    "delivery_message": "Product Delivered On Time"
  }, ...],
  ...
}
```

GET /dcs/(*dc_id*)/routes/import-status/(*txid*)

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **txid** (*string*) – The transaction id returned by the add route API call

- Status Codes:**
- **202 Accepted** – The **route** is still being processed.
 - **400 Bad Request** – **Route** import failed. See the response for detail.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.
 - **404 Not Found** – This transaction id does not exist.

Status 400 error codes

Error Code	Description
3000	Import error. See the 'errors' field for detail.

Example pending response

```
{
  "status": "pending",
  "data": {}
}
```

Example success response

```
{
  "status": "success",
  "data": {
    "route_id": "the route id"
  }
}
```

Example error response

```
{
  "status": "error",
  "data": {
    "message": "An error occurred. See 'errors' field for detail",
    "code": 3000,
    "errors": [{
      "message": "Error description"
    }]
  }
}
```

GET /dcs/(*dc_id*)/routes/find_by_date/(*date*)

- Parameters:
- *dc_id* (*string*) – The dc id
 - *date* (*string*) – The date in YYYY-MM-DD format

- Status Codes:
- 200 OK – Route is returned.
 - 401 Unauthorized – API key is invalid.
 - 403 Forbidden – DC does not exist or user does not have permission to access it.

Example response

```
{ "routes":  
  [  
    {  
      "status": "success",  
      "data": {  
        "route": {  
          "id": "globally unique route id",  
          "name": "route name",  
          "start_time": 1510696728000,  
          "waypoint_ids": ["wpA", "wpB"],  
          "start_warehouse_id": "warehouse id",  
          "end_warehouse_id": "warehouse id",  
          "assigned_driver_id": "driver id",  
          "vehicle_id": "vehicle id",  
          "version": 12,  
          "is_active": True,  
          "is_finalized": False,  
          "imported_timestamp": 1510696728000,  
          "started_timestamp": 1510696728000,  
          "finalized_timestamp": 1510696728000  
        }  
      }  
    },  
    ...  
  ]  
}
```

PATCH /dcs/(*dc_id*)/routes/(*route_id*)/vehicle-assignment

- Parameters:
- *dc_id* (*string*) – The dc id
 - *route_id* (*string*) – The route id

Request JSON Object:

- **vehicle_id** (*string*) – The vehicle's id. If the vehicle does not exist, it will be created automatically.

- Status Codes:**
- **200 OK** – **Route** was assigned.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.

Example request

```
{  
  "vehicle_id": "Vehicle's id"  
}
```

PUT /dcs/(**dc_id**)/routes/(**route_id**)/vehicle-assignment

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id

Request JSON Object:

- **vehicle_id** (*string*) – The vehicle's id. If the vehicle does not exist, it will be created automatically.

- Status Codes:**
- **200 OK** – **Route** was assigned.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.

Example request

```
{  
  "vehicle_id": "Vehicle's id"  
}
```

PATCH /dcs/(*dc_id*)/routes/(*route_id*)/driver-assignment

- Parameters:
- *dc_id* (*string*) – The dc id
 - *route_id* (*string*) – The route id

Request JSON Object:

- *driver_id* (*string*) – The driver's id

- Status Codes:
- 200 OK – Route was assigned.
 - 401 Unauthorized – API key is invalid.
 - 403 Forbidden – DC does not exist or user does not have permission to access it.
 - 404 Not Found – The *route* or *driver* do not exist.

Example request

```
{  
  "driver_id": "John's id"  
}
```

PUT /dcs/(*dc_id*)/routes/(*route_id*)/driver-assignment

- Parameters:
- *dc_id* (*string*) – The dc id
 - *route_id* (*string*) – The route id

Request JSON Object:

- **driver_id** (*string*) – The driver's id

- Status Codes:**
- **200 OK** – **Route** was assigned.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.
 - **404 Not Found** – The **route** or **driver** do not exist.

Example request

```
{  
  "driver_id": "John's id"  
}
```

DELETE /dcs/(**dc_id**)/routes/(**route_id**)/driver-assignment

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id

- Status Codes:**
- **200 OK** – **Route** assignment was removed.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.
 - **404 Not Found** – **Route** does not exist.

GET /dcs/(**dc_id**)/routes/(**route_id**)/completion-time

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id

- Status Codes:**
- 200 OK – [Route Completion Time](#) is returned.
 - 401 Unauthorized – API key is invalid.
 - 403 Forbidden – [DC](#) does not exist or user does not have permission to access it.
 - 404 Not Found – [Route](#) does not exist.

Example response

```
{
  "status": "success",
  "data": {
    "completion_time": {
      "timestamp": 1510696728000,
      "type": "ESTIMATED"
    }
  }
}
```

PATCH /dcs/[\(dc_id\)](#)/routes/[\(route_id\)](#)/deactivate

- Parameters:**
- `dc_id` (*string*) – The dc id
 - `route_id` (*string*) – The route id

- Status Codes:**
- 200 OK – [Route](#) was deactivated.
 - 401 Unauthorized – API key is invalid.
 - 403 Forbidden – [DC](#) does not exist or user does not have permission to access it.
 - 404 Not Found – [Route](#) does not exist.

PUT /dcs/[\(dc_id\)](#)/routes/[\(route_id\)](#)/deactivate

- Parameters:**
- `dc_id` (*string*) – The dc id
 - `route_id` (*string*) – The route id

- Status Codes:**
- [200 OK](#) – [Route](#) was deactivated.
 - [401 Unauthorized](#) – API key is invalid.
 - [403 Forbidden](#) – [DC](#) does not exist or user does not have permission to access it.
 - [404 Not Found](#) – [Route](#) does not exist.

GET `/dcs/(dc\_id)/routes/(route\_id)/waypoints/(waypoint\_id)`

- Parameters:**
- `dc_id` (*string*) – The dc id
 - `route_id` (*string*) – The route id
 - `waypoint_id` (*string*) – The waypoint id

- Status Codes:**
- [200 OK](#) – Waypoint objects are returned.
 - [401 Unauthorized](#) – API key is invalid.
 - [403 Forbidden](#) – [DC](#) does not exist or user does not have permission to access it.
 - [404 Not Found](#) – Waypoint does not exist.

Example Responses

```
{
  "waypoint": {
    "id": "the-waypoint-id",
    "customer_id": "the-customer-id",
    "status": "COMPLETED",
    "completed_timestamp": null,
    "estimated_time_of_arrival": 1570561955123,
    "waiting_time_seconds": 0,
    "waypoints_ahead": 1
  }
}
```

Please note that the *completed_timestamp*, *estimated_time_of_arrival* and *waypoints_ahead* fields may be null.

- If the Waypoint status is *PENDING*, the *completed_timestamp* will be null.

- If the Waypoint status is *COMPLETED*, the *estimated_time_of_arrival* and *waypoints_ahead* will be null.
- The *estimated_time_of_arrival* may also be null in other situations. If the Waypoint's location is unknown, or the route was just imported, the Waypoint will not have an *estimated_time_of_arrival*.
- If the Waypoint is next in line to be visited, *waypoints_ahead* will be 0.

POST /dcs/(*dc_id*)/routes/(*route_id*)/waypoints

- Parameters:
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id

- Status Codes:
- **200 OK** – Waypoint was added.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.
 - **404 Not Found** – **Route** does not exist.

Status 400 error codes

Error Code	Description
3006	Cannot import another waypoint with the same id.
3007	Cannot import another delivery with the same id.

Example request

```
{
  "waypoint": {
    "id": "globally unique waypoint id",
    "customer_id": "customer id",
    "name": "waypoint name",
    "address": "physical address",
    "location": {
      "latitude": 1.0,
      "longitude": 2.0
    },
    "service_time_minutes": 999,
    "time_windows": [{
      "start": 1510864915000,
      "end": 1510868515000
    }],
    "deliveries": [{
      "id": "globally unique delivery id",
      "product": "product",
      "quantity": 4355.0
    }]
  }
}
```

GET /dcs/([dc_id](#))/routes/([route_id](#))/waypoints

Parameters:

- **dc_id** (*string*) – The dc id
- **route_id** (*string*) – The route id

Status Codes:

- [200 OK](#) – Waypoint objects are returned.
- [401 Unauthorized](#) – API key is invalid.
- [403 Forbidden](#) – [DC](#) does not exist or user does not have permission to access it.
- [404 Not Found](#) – [Route](#) does not exist.

Example Responses

```
{
  "waypoints": [{
    "id": "the-waypoint-id",
    "customer_id": "the-customer-id",
    "status": "COMPLETED",
    "completed_timestamp": null,
    "estimated_time_of_arrival": 1570561955123,
    "waypoints_ahead": 1,
    "waiting_time_seconds": 0,
  }, ...],
}
```

Please note that the *completed_timestamp*, *estimated_time_of_arrival* and *waypoints_ahead* fields may be null.

- If the Waypoint status is *PENDING*, the *completed_timestamp* will be null.
- If the Waypoint status is *COMPLETED*, the *estimated_time_of_arrival* and *waypoints_ahead* will be null.
- The *estimated_time_of_arrival* may also be null in other situations. If the Waypoint's location is unknown, or the route was just imported, the Waypoint will not have an *estimated_time_of_arrival*.
- If the Waypoint is next in line to be visited, *waypoints_ahead* will be 0.

PATCH /dcs/(*dc_id*)/routes/(*route_id*)/activate

Parameters:

- *dc_id* (*string*) – The dc id
- *route_id* (*string*) – The route id

Status Codes:

- 200 OK – Route was activated.
- 401 Unauthorized – API key is invalid.
- 403 Forbidden – DC does not exist or user does not have permission to access it.
- 404 Not Found – Route does not exist.

PUT /dcs/(*dc_id*)/routes/(*route_id*)/activate

- Parameters:
- *dc_id* (*string*) – The dc id
 - *route_id* (*string*) – The route id

- Status Codes:
- 200 OK – *Route* was activated.
 - 401 Unauthorized – API key is invalid.
 - 403 Forbidden – *DC* does not exist or user does not have permission to access it.
 - 404 Not Found – *Route* does not exist.

PATCH /dcs/(*dc_id*)/routes/(*route_id*)/finish

- Parameters:
- *dc_id* (*string*) – The dc id
 - *route_id* (*string*) – The route id

- Status Codes:
- 200 OK – *:Route* was finished.
 - 401 Unauthorized – API key is invalid.
 - 403 Forbidden – *DC* does not exist or user does not have permission to access it.
 - 404 Not Found – *Route* does not exist.

Example Responses Non-errored responses will include the route id and one of the following messages: - *ALREADY_FINISHED* if the route was already finished before calling the endpoint - *SUCCESS* if the route was active when calling the endpoint and successfully was finished

```
{
  "route_id": "a route id",
  "message": "ALREADY_FINISHED"
}
```

```
{
  "route_id": "a route id",
  "message": "SUCCESS"
}
```

PUT /dcs/(*dc_id*)/routes/(*route_id*)/finish

- Parameters:
- *dc_id* (*string*) – The dc id
 - *route_id* (*string*) – The route id

- Status Codes:
- 200 OK – :Route was finished.
 - 401 Unauthorized – API key is invalid.
 - 403 Forbidden – DC does not exist or user does not have permission to access it.
 - 404 Not Found – Route does not exist.

Example Responses Non-errored responses will include the route id and one of the following messages: - *ALREADY_FINISHED* if the route was already finished before calling the endpoint - *SUCCESS* if the route was active when calling the endpoint and successfully was finished

```
{
  "route_id": "a route id",
  "message": "ALREADY_FINISHED"
}
```

```
{
  "route_id": "a route id",
  "message": "SUCCESS"
}
```

PATCH /dcs/(*dc_id*)/routes/(*route_id*)/start

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id

- Status Codes:**
- **200 OK** – :Route was started (or already started).
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – DC does not exist or user does not have permission to access it.
 - **404 Not Found** – Route does not exist.

Example Responses Non-errored responses will include the route id and one of the following messages: - *ALREADY_STARTED* if the route was already started before calling the endpoint - *SUCCESS* if the route was not yet started when calling then endpoint and was successfully started - 'NO_DRIVER_ASSIGNED' if the route could not be started because no driver was assigned

```
{
  "route_id": "a route id",
  "message": "ALREADY_STARTED"
}
```

```
{
  "route_id": "a route id",
  "message": "SUCCESS"
}
```

```
{
  "route_id": "a route id",
  "message": "NO_DRIVER_ASSIGNED"
}
```

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id

- Status Codes:**
- **200 OK** – :Route was started (or already started).
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – DC does not exist or user does not have permission to access it.
 - **404 Not Found** – Route does not exist.

Example Responses Non-errored responses will include the route id and one of the following messages: - *ALREADY_STARTED* if the route was already started before calling the endpoint - *SUCCESS* if the route was not yet started when calling then endpoint and was successfully started - 'NO_DRIVER_ASSIGNED' if the route could not be started because no driver was assigned

```
{
  "route_id": "a route id",
  "message": "ALREADY_STARTED"
}
```

```
{
  "route_id": "a route id",
  "message": "SUCCESS"
}
```

```
{
  "route_id": "a route id",
  "message": "NO_DRIVER_ASSIGNED"
}
```

Note: Routes that have been analyzed or are active cannot be deleted.

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id

- Status Codes:**
- **200 OK** – Route was deleted.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – DC does not exist or user does not have permission to access it.
 - **404 Not Found** – Route does not exist.

Status 400 error codes

Error Code	Description
3004	Cannot delete already analyzed routes.
3010	Cannot delete active routes.

GET /dcs/(*dc_id*)/routes/(*route_id*)

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id

- Status Codes:**
- **200 OK** – Route is returned.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – DC does not exist or user does not have permission to access it.
 - **404 Not Found** – Route does not exist.

Example response

```
{
  "status": "success",
  "data": {
    "route": {
      "id": "globally unique route id",
      "name": "route name",
      "start_time": 1510696728000,
      "waypoint_ids": ["wpA", "wpB"],
      "start_warehouse_id": "warehouse id",
      "end_warehouse_id": "warehouse id",
      "assigned_driver_id": "driver id",
      "vehicle_id": "vehicle id",
      "version": 12,
      "is_active": true,
      "is_finalized": false,
      "imported_timestamp": 1510696728000,
      "started_timestamp": 1510696728000,
      "finalized_timestamp": 1510696728000
    }
  }
}
```

POST /dcs/([dc_id](#))/routes

This endpoint is asynchronous and will return a transaction id to be used to check the route import status.

Parameters:

- `dc_id` (*string*) – The dc id

Request JSON Object:

- `route` (*json*) – The route to be imported. See [import route model](#)

Status Codes:

- [202 Accepted](#) – The request to import a [Route](#) has been accepted. The returned *txid* should be used to poll for route import status.
- [400 Bad Request](#) – Check error code within the response.
- [401 Unauthorized](#) – API key is invalid.
- [403 Forbidden](#) – [DC](#) does not exist or user does not have permission to access it.

Status 400 error codes

Error Code	Description
3005	Cannot import another route with the same id.
3006	Cannot import another waypoint with the same id.
3007	Cannot import another delivery with the same id.

Example request

```

{
  "route": {
    "id": "globally unique route id",
    "name": "route name",
    "driver_id": "globally unique driver id",
    "start_time": 1510860915000,
    "activate_after_successful_import": true,
    "start_warehouse": {
      "id": "warehouse id",
      "name": "warehouse name",
      "address": "physical address",
      "location": {
        "latitude": 3.0,
        "longitude": 4.0
      }
    },
    "end_warehouse": {
      "id": "warehouse id",
      "name": "warehouse name",
      "address": "physical address",
      "location": {
        "latitude": 3.0,
        "longitude": 4.0
      }
    },
    "vehicle": {
      "id": "globally unique vehicle id"
    },
    "waypoints": [{
      "id": "globally unique waypoint id",
      "customer_id": "customer id",
      "name": "waypoint name",
      "address": "physical address",
      "contact_name": "John Smith",
      "contact_phone": "555-555-5555",
      "contact_email": "example-email@foxtrot.io",
      "location": {
        "latitude": 1.0,
        "longitude": 2.0
      },
      "service_time_minutes": 999,
      "time_windows": [{
        "start": 1510864915000,
        "end": 1510868515000
      }],
      "deliveries": [{
        "id": "globally unique delivery id",
        "product": "product",
        "quantity": 4355.0
      }]
    }],
    {
      "id": "globally unique waypoint id 2",

```



```
    "customer_id": "customer id",
    "name": "waypoint name",
    "address": "physical address",
    "location": {
      "latitude": 3.0,
      "longitude": 4.0
    },
    "service_time_minutes": 999,
    "time_windows": [],
    "deliveries": [{
      "id": "delivery B",
      "product": "product",
      "quantity": 342.0
    }]
  }
]
}
```

Example successful response

```
{
  "status": "success",
  "data": {
    "txid": "the transaction id"
  }
}
```

Example error response

```
{
  "data": {
    "code": 3006,
    "message": "Cannot import another waypoint with the same id"
  },
  "status": "error"
}
```

Invoiced Routes API

Summary

Resource	Operation	Description
Invoiced Routes	POST /dcs/(dc_id)/invoiced-routes	Add an importable invoiced route

Details

POST /dcs/([dc_id](#))/invoiced-routes

This endpoint is asynchronous and will return a transaction id to be used to check the route import status.

Parameters:

- **dc_id** (*string*) – The dc id

Request JSON Object:

- **route** (*json*) – The invoiced route to be imported. See [import invoiced route model](#)

Status Codes:

- [202 Accepted](#) – The request to import an [Invoiced Route](#) has been accepted. The returned *txid* should be used to poll for the route import status.
- [400 Bad Request](#) – Check error code within the response.
- [401 Unauthorized](#) – API key is invalid.
- [403 Forbidden](#) – [DC](#) does not exist or user does not have permission to access it.

Status 400 error codes

Error Code	Description
3005	Cannot import another route with the same id.
3006	Cannot import another waypoint with the same id.
3008	Cannot import another invoice with the same id.
3009	Cannot import another invoice item with the same id.

Example request

```
{
  "route": {
    "id": "globally unique route id",
    "name": "name",
    "driver_id": "driver id",
    "start_time": 1510860915000,
    "activate_after_successful_import": true,
    "start_warehouse": {
      "id": "warehouse id",
      "name": "warehouse name",
      "address": "physical address",
      "location": {
        "latitude": 3.0,
        "longitude": 4.0
      }
    },
    "end_warehouse": {
      "id": "warehouse id",
      "name": "warehouse name",
      "address": "physical address",
      "location": {
        "latitude": 3.0,
        "longitude": 4.0
      }
    },
    "vehicle": {
      "id": "globally unique vehicle id"
    },
    "waypoints": [{
      "id": "globally unique waypoint id",
      "customer_id": "customer id",
      "name": "waypoint name",
      "address": "physical address",
      "location": {
        "latitude": 1.0,
        "longitude": 2.0
      }
    },
    "service_time_minutes": 999,
    "time_windows": [{
      "start": 1510864915000,
      "end": 1510868515000
    }],
    "invoices": [{
      "id": "invoice A",
      "payment_method": "CASH",
      "invoice_name_attributes": [{"key": "key1", "value": "value2"}],
      "items": [{
        "id": "item A",
        "sku": "sku A",
        "name": "name A",
        "quantity": 123.0,
        "display_unit_of_measure": "bottles",
        "total_price_cents": 9999,

```

```

        "currency": "USD"
      }]
    },
    {
      "id": "globally unique waypoint id",
      "customer_id": "customer id",
      "name": "waypoint name",
      "address": "physical address",
      "location": {
        "latitude": 3.0,
        "longitude": 4.0
      },
      "service_time_minutes": 999,
      "time_windows": [],
      "invoices": [{
        "id": "invoice B",
        "payment_method": "CASH",
        "invoice_name_attributes": [{"key": "key1", "value": "value2"}],
        "items": [{
          "id": "item B",
          "sku": "sku B",
          "name": "name B",
          "quantity": 123.0,
          "display_unit_of_measure": "cases",
          "total_price_cents": 9999,
          "currency": "USD"
        }]
      }]
    }
  ]
}

```

Example successful response

```

{
  "status": "success",
  "data": {
    "txid": "the transaction id"
  }
}

```

Example error response

```
{  
  "data": {  
    "code": 3006,  
    "message": "Cannot import another waypoint with the same id"  
  },  
  "status": "error"  
}
```

Invoices API

Summary

Resource	Operation	Description
Invoice Status	GET /dcs/(dc_id)/routes/(route_id)/invoices	Get all Invoice status for the route

Details

GET /dcs/([dc_id](#))/routes/([route_id](#))/invoices

- Parameters:
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id

- Status Codes:
- [200 OK](#) – [Invoice Status](#) was returned.
 - [400 Bad Request](#) – Check error code within the response.
 - [401 Unauthorized](#) – API key is invalid.
 - [403 Forbidden](#) – [DC](#) does not exist or user does not have permission to access it.

Error Code	Description
3003	Route with that id does not exist.

Example response

```
{
  "status": "success",
  "data": {
    "invoices": [{
      "id": "globally unique invoice id",
      "payment_method": "CASH",
      "invoice_name_attributes": [{"key": "Customer TAX ID", "value": "A-1232-001"}],
      "items": [{
        "id": "globally unique item id",
        "sku": "sku A",
        "name": "name A",
        "quantity": 123.0,
        "total_price_cents": 9999,
        "currency": "USD",
        "status": {
          "status": "SUCCESSFUL",
          "timestamp": 1531267636897
        }
      }]
    }], {
      "id": "globally unique invoice id 2",
      "payment_method": "CASH",
      "invoice_name_attributes": [{"key": "Customer TAX ID", "value": "A-1232-001"}],
      "items": [{
        "id": "globally unique item id 2",
        "sku": "sku B",
        "name": "name B",
        "quantity": 123.0,
        "total_price_cents": 9999,
        "currency": "USD",
        "status": {
          "status": "FAILED",
          "timestamp": 1531267636897,
          "reason": {
            "code": "00001",
            "message": "Customer closed"
          }
        }
      }]
    }
  ]
}
```


Routes Accounting API

Summary

Resource	Operation	Description
Route Accounting Record	GET /dcs/(dc_id)/routes/(route_id)/accounting	Get all records for the route

Details

GET [/dcs/\(dc_id\)/routes/\(route_id\)/accounting](#)

- Parameters:
- **dc_id** (*string*) – The dc id
 - **route_id** (*string*) – The route id

- Status Codes:
- [200 OK](#) – [Accounting Record](#) was returned.
 - [400 Bad Request](#) – Check error code within the response.
 - [401 Unauthorized](#) – API key is invalid.
 - [403 Forbidden](#) – [DC](#) does not exist or user does not have permission to access it.

Error Code	Description
3003	Route with that id does not exist.

Example response

```
{  
  "status": "success",  
  "data": {  
    "records": [{  
      "id": "record A",  
      "driver_id": "driver A",  
      "cash_received_cents": 9999,  
      "cash_container_id": "bag A"  
    }]  
  }  
}
```

Analytics API

Summary

This API is used to download Foxtrot data programatically so you can feed it into your different tools. For help and definitions of each of the columns offered please refer to the File Download Page using your Foxtrot account.

Accessing Foxtrot data requires two steps, one is to create a request to download. This is where you define what dates, Distribution Centers and Analysis types you want.

There are two types of downloadable data:

- **ROUTE_ANALYTICS:** These files contain per route analysis results
- **ATTEMPT_ANALYTICS:** These files contain per attempt analysis results

Once a request is created, you can get a list of the files generated using the transaction ID returned on the request creation.

Resource	Operation	Description
Analytics	GET /dcs/(dc_id)/routes/(route_id)/analytics/attempts	Get Attempt Analyt
	GET /dcs/(dc_id)/routes/(route_id)/analytics/route	Get Route Analytics
File Downloads	GET /analytics/column-definitions	Get Analysis Colum
	POST /analytics/download-requests	Post Analytics Dow

Resource	Operation	Description
	GET /analytics/download-requests/(download_request_txid)/files	Get Route Analytics

Details 🔗

GET /analytics/column-definitions

- Status Codes:
- [200 OK](#) – list of column definitions.
 - [401 Unauthorized](#) – API key is invalid.

Example response

```
{
  "status": "success",
  "data": {
    "route_analysis_column_definitions": {
      "dc_id": {
        "column_name": "DC ID",
        "column_description": "Distribution Center ID"
      },
      "dc_time_zone": {
        "column_name": "DC Time Zone",
        "column_description": "Local DC time zone"
      }, ...
    },
    "attempt_analysis_column_definitions": {
      "dc_id": {
        "column_name": "DC ID",
        "column_description": "Distribution Center ID"
      },
      "waypoint_id": {
        "column_name": "Waypoint ID",
        "column_description": "The unique identifier for this customer in this route. This
is usually used by systems to identify a certain instance of a customer in a route."
      }, ...
    }
  }
}
```

POST /analytics/download-requests

- Parameters:**
- **dc_ids** (*list*) – List of distribution center IDs, i.e. ["dc_id_1", "dc_id_2"]
 - **from_date** (*string*) – The start date of date range you want to download files for, in YYYY-MM-DD format, local DC time for each DC
 - **to_date** (*string*) – The end date of date range you want to download files for, in YYYY-MM-DD format, local DC time for each DC
 - **analytics_types** (*list*) – List containing ROUTE_ANALYTICS and/or ATTEMPT_ANALYTICS, i.e. ["ROUTE_ANALYTICS", "ATTEMPT_ANALYTICS"]

- Status Codes:**
- **200 OK** – An object containing your download request transaction id.
 - **400 Bad Request** – Check error code within the response.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – One of the DC IDs specified does not belong to the account API Key.

Example response

```
{
  "status": "success",
  "data": {
    "download_request_txid": "abcdef123456"
  }
}
```

GET /dcs/(dc_id)/routes/(route_id)/analytics/attempts

- Status Codes:**
- [200 OK](#) – An object with the attempt analysis results.
 - [401 Unauthorized](#) – API key is invalid.
 - [403 Forbidden](#) – The DC ID specified does not belong to the account API Key.
 - [404 Not Found](#) – The Route ID is invalid or the Attempt Analytics are not yet available

Example response

```
{
  "status": "success",
  "data": {
    "attempt_analytics": {
      "dc_id": "2123",
      "dc_name": "My DC",
      ...
    }, ...
  }
}
```

GET `/dcs/(<dc_id>)/routes/(<route_id>)/analytics/route`

- Status Codes:**
- [200 OK](#) – An object with the route analysis results.
 - [401 Unauthorized](#) – API key is invalid.
 - [403 Forbidden](#) – The DC ID specified does not belong to the account API Key.
 - [404 Not Found](#) – The Route ID is invalid or the Route Analytics are not yet available

Example response

```
{
  "status": "success",
  "data": {
    "route_analytics": [{
      "dc_id": "2123",
      "dc_name": "My DC",
      ...
    }, ...]
  }
}
```

GET /analytics/download-requests/*(download_request_txid)***/files**

Parameters:

- **download_request_txid** (*string*) – The transaction ID returned when creating an analytics download request.

Status Codes:

- [200 OK](#) – list of object(s) containing file data.
- [202 Accepted](#) – files still pending generation, returning the same download request text id.
- [400 Bad Request](#) – Check error code within the response.
- [401 Unauthorized](#) – API key is invalid.

Error Code	Description
6000	Invalid Request Transaction ID

Example response

```
{
  "status": "success",
  "data": {
    "download_request_files": [
      {
        "file_type": "CSV",
        "download_link": "https://foxtrotsystems.com/files/5d35htaa04a8f942f7",
        "expiration_timestamp": 1483287694000,
        "download_type": "ROUTE_ANALYTICS",
      }, ...
    ]
  }
}
```

Property	Type	Nullable	Description
file_type	string	false	File type, i.e. CSV
download_link	string	false	Link to download files
expiration_timestamp	string	false	Expiration date for files
download_type	string	false	Data that you are downloading, i.e. Attempt Analytics



Response FAQ

Why am I getting back a *download_request_txid*?

If you receive your *download_request_txid* as a response, this means the files are still being generated.

Customers API

Summary

Resource	Operation	Description
	GET /dcs/(dc_id)/customers/(customer_id)/waypoints	Get waypoint information f
Customers	GET /dcs/(dc_id)/customers/(customer_id)/confident-location	Get the confident location i
	GET /dcs/(dc_id)/customers/(customer_id)	Get a customer

Details

GET /dcs/([dc_id](#))/customers/([customer_id](#))/confident-location

- Parameters:
- **dc_id** (*string*) – The dc id
 - **customer_id** (*string*) – The customer's id

- Status Codes:
- [200 OK](#) – A JSON object with the customer's location and id is returned
 - [400 Bad Request](#) – Check error code within the response.
 - [401 Unauthorized](#) – API key is invalid.
 - [403 Forbidden](#) – **DC** does not exist or user does not have permission to access it.
 - [404 Not Found](#) – Customer does not exist.

Error Code	Description
7001	Customer location not confident.

Example request

```
{  
  "dc_id": "Foxtrot-DC",  
  "customer_id": "johns-burger-shack"  
}
```

Example response

```
{  
  "status": "success",  
  "data": {  
    "customer_id": "johns-burger-shack",  
    "latitude": 1.00456,  
    "longitude": -3.123534  
  }  
}
```

GET /dcs/*(dc_id)*/customers/*(customer_id)*/waypoints

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **customer_id** (*string*) – The customer's id

- Status Codes:**
- **200 OK** – waypoints are returned.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – DC does not exist or user does not have permission to access it.

Example response

```
{
  "waypoints": [
    {
      "id": "The waypoint's id",
      "customer_id": "The customer's id",
      "status": "PENDING",
      "completed_timestamp": null,
      "estimated_time_of_arrival": 123456,
      "waypoints_ahead": 4,
      "waiting_time_seconds": 600,
      "route_id": "The route's id"
    },
    ...
  ]
}
```

Note that this only returns the most recent 10 waypoints for the customer.

GET /dcs/(*dc_id*)/customers/(*customer_id*)

- Parameters:**
- **dc_id** (*string*) – The dc id
 - **customer_id** (*string*) – The customer's id

- Status Codes:**
- **200 OK** – Customer object is returned.
 - **401 Unauthorized** – API key is invalid.
 - **403 Forbidden** – **DC** does not exist or user does not have permission to access it.
 - **404 Not Found** – Customer does not exist.

Example response

```
{
  "status": "success",
  "data": {
    "customer": {
      "id": "the customer's id",
      "name": "the customer's name",
      "imported_address": "address string",
      "imported_location": {
        "latitude": 37.77493,
        "longitude": 122.41942
      },
      "inferred_location": {
        "latitude": 37.77483,
        "longitude": 122.41940
      }
    }
  }
}
```

Note that the location object may be null if Foxtrot does not have an imported location for the customer.

Webhooks API

Summary

Resource	Operation	Description
Webhooks	GET /webhooks	Gets all registered Webhooks
	POST /webhooks	Creates a Webhook
	PATCH /webhooks/(id)	Updates a Webhook
	PUT /webhooks/(id)	
	DELETE /webhooks/(id)	Deletes a Webhook

Details

GET /webhooks

- Status Codes:
- [200 OK](#) – The [DC](#) objects are returned.
 - [401 Unauthorized](#) – API key is invalid.

Example response

```
{
  "status": "success",
  "data": {
    "webhooks": [{
      "id": "the-webhook-identifier",
      "url": "https://example.com/webhook",
      "enabled": true,
      "dc_ids": ["1", "2", "3"],
      "events": ["ROUTE_CREATED", "ROUTE_STARTED", "ROUTE_ETAS_CHANGED",
"WAYPOINT_STATUS_CHANGED"]
    }, ...]
  }
}
```

POST /webhooks

- Parameters:**
- `webhook_url` (*string*) – The Webhook URL to register
 - `events` (*list*) – The event types that should be sent
 - `dc_ids` (*list*) – The [DCs](#) for which events should be sent

- Status Codes:**
- [200 OK](#) – The newly created [Webhook](#) object is returned.
 - [400 Bad Request](#) – One or more parameters was invalid. Check error code within the response.
 - [401 Unauthorized](#) – API key is invalid.

Example response

```
{
  "status": "success",
  "data": {
    "webhook": {
      "id": "the-webhook-identifier",
      "url": "https://example.com/webhook",
      "enabled": true,
      "dc_ids": ["1", "2", "3"],
      "events": ["ROUTE_CREATED", "ROUTE_STARTED"]
    }
  }
}
```

Note that Webhook URLs must be fully formed URLs using the http or https schema, e.g. <https://example.com/my-webhook>.

You can register for multiple event-types with one Webhook URL. See [WebhookEventType](#) for a complete description of all the available webhook events you can register for.

PATCH /webhooks/*(id)*

- Parameters:**
- **id** (*string*) – The webhook id
 - **enabled** (*bool*) – Whether this webhook should be disabled or disabled
- Status Codes:**
- **200 OK** – The updated [Webhook](#) object is returned.
 - **400 Bad Request** – One or more parameters was invalid. Check error code within the response.
 - **401 Unauthorized** – API key is invalid.
 - **404 Not Found** – The Webhook with this ID could not be found.

Example response

```
{
  "status": "success",
  "data": {
    "webhook": {
      "id": "the-webhook-identifier",
      "url": "https://example.com/webhook",
      "enabled": true,
      "dc_ids": ["1", "2", "3"],
      "events": ["ROUTE_CREATED", "ROUTE_STARTED"]
    }
  }
}
```

Note that currently, only the enabled/disabled status of the Webhook can be modified. If you need to modify other properties of the Webhook, you will need to delete the existing Webhook and add another.

PUT /webhooks/*(id)*

- Parameters:**
- **id** (*string*) – The webhook id
 - **enabled** (*bool*) – Whether this webhook should be disabled or disabled
- Status Codes:**
- **200 OK** – The updated [Webhook](#) object is returned.
 - **400 Bad Request** – One or more parameters was invalid. Check error code within the response.
 - **401 Unauthorized** – API key is invalid.
 - **404 Not Found** – The Webhook with this ID could not be found.

Example response

```
{
  "status": "success",
  "data": {
    "webhook": {
      "id": "the-webhook-identifier",
      "url": "https://example.com/webhook",
      "enabled": true,
      "dc_ids": ["1", "2", "3"],
      "events": ["ROUTE_CREATED", "ROUTE_STARTED"]
    }
  }
}
```

Note that currently, only the enabled/disabled status of the Webhook can be modified. If you need to modify other properties of the Webhook, you will need to delete the existing Webhook and add another.

DELETE /webhooks/*(id)*

Parameters:

- **id** (*string*) – The Webhook id

Status Codes:

- **200 OK** – The Webhook was deleted.
- **404 Not Found** – The Webhook with this ID could not be found.